

Quiz Questions:

Question 1:

What is the output of the following code?

```
my_list = [10, 20, 30, 40, 50]
print(my_list[1:4])
```

- a) [20, 30, 40]
- b) [10, 20, 30]
- c) [30, 40, 50]
- d) [20, 40, 50]

Question 2:

Which method is used to add multiple elements to the end of a list?

- a) `add()`
- b) `append()`
- c) `extend()`
- d) `insert()`

Question 3:

Consider the following list:

```
fruits = ['apple', 'banana', 'orange']
```

How can you remove 'banana' from the list?

- a) `fruits.remove('banana')`
- b) `fruits.delete('banana')`
- c) `fruits.pop('banana')`
- d) `fruits.exclude('banana')`

Question 4:

What does the `len()` function return when applied to a list?

- a) The sum of all elements in the list
- b) The average of all elements in the list
- c) The number of elements in the list
- d) The maximum element in the list

Question 5:

Which of the following list comprehensions generates a list of even numbers from 0 to 10?

- a) `[x for x in range(11) if x % 2 == 0]`
- b) `[x for x in range(10) if x % 2 == 0]`
- c) `[x**2 for x in range(11)]`
- d) `[x**2 for x in range(10) if x % 2 == 0]`

Task 1:

Reverse List:

Write Python code to reverse the order of elements in the given list `my_list`.
Print the reversed list.

```
my_list = [10, 20, 30, 40, 50, 11]

# Your code here

# Output should be: [11,50,40,30,20,10]
```

Task 2:

Common Elements:

Given two lists `list1` and `list2`, find and print the common elements between them.

```
list1 = [1, 2, 3, 4, 5]
list2 = [4, 5, 6, 7, 8]

# Your code here
```

```
# Output should be: [4, 5]
```

Task 3:

Unique Elements:

Create a new list `unique_list` containing only the unique elements from the given list `original_list`. Print the unique list.

```
original_list = [1, 2, 2, 3, 4, 4, 5]

# Your code here

# Output should be: [1, 2, 3, 4, 5]
```

Task 4:

Remove Duplicates:

Remove duplicate elements from the given list `duplicated_list` and print the list without duplicates while preserving the order.

```
duplicated_list = [1, 2, 2, 3, 4, 4, 5]

# Your code here

# Output should be: [1, 2, 3, 4, 5]
```

Exercise 1: List Concatenation

Write a Python script that concatenates two lists and prints the result.

Exercise 2: List Repetition

Write a Python script that repeats a list three times and prints the result.

Exercise 3: List Removal

Write a Python script that removes the elements at even indices from a list.

Exercise 4: List Insertion

Write a Python script that inserts the numbers 10, 11, and 12 at the beginning of a list

List comprehensions

1. **Square Numbers:** Create a list of squares of numbers from 1 to 10.
2. **Even Numbers:** Generate a list of even numbers from 1 to 20.
3. **Words Lengths:** Given a list of words, create a list containing the lengths of each word.

```
words = ["apple", "banana", "cherry", "date"]
```